Truth as a Compression Artifact in Language Model Training

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Abstract

When a transformer is trained from scratch on corpora that solve the same mathematical problem both correctly and incorrectly, which solution does it prefer? The usual test is pairwise, and it is not neutral: when the error can take several forms, a model that merely reproduces training frequencies wins it without preferring anything. We instead score the correct completion against every string the error generator can produce, a share on which frequency matching scores exactly 0.500. At 86M parameters the correct completion holds $59.0\% \pm 0.8\%$ of the mass against random errors (59.4% on an independent corpus), where the single realized error gives 77.6% and pair accuracy 87.4%, a three- to fourfold overstatement of the excess above chance; below 86M the primary corpus shows no preference. The coherent false rules of the main corpora are not filtered at any size (43.5–48.8%; the shortfall below 0.5 is a completion-length effect). Two experiments built to identify a compression mechanism raise the pairwise measures while the summed share stays flat or falls: both are negative controls, and compression remains motivation, not a measured mechanism. With the scored false strings held fixed, a false rule whose selector the model can learn moves the correct share from 5 points above the null to 7 below (418M–3.08B), though the correct completion still outweighs the selected string. What these models leave unfiltered is consistent or learnable false structure (one single-rule corruption excepted), not falsehood as such.

1 Introduction

Training corpora are noisy: the same question receives contradictory answers in different documents, and yet language models trained on such data tend to prefer the correct one. Without an account of *why* this filtering works we cannot predict *when* it fails, for instance on internally consistent errors.

Scaling [Kadavath et al., 2022], frequency [Kandpal et al., 2023] and truth-correlated representations [Marks and Tegmark, 2024] each predict factual accuracy, and calibrated models must hallucinate at a rate set by the corpus [Kalai and Vempala, 2024], but none of this says why the training objective alone would favor one answer over another at equal frequency and format. Minimizing cross-entropy minimizes code length [Shannon, 1948, Delétang et al., 2024], which links training to the Minimum Description Length principle [Rissanen, 1978, Grünwald, 2007] and suggests that an error the model cannot predict from the problem is harder to accommodate than one it can, so a *coherent* false system would go unfiltered. This framing is motivation only: both of our attempts to establish it as the operative mechanism turn out to measure something else.

The pairwise measure used throughout the literature on contradictory data is not neutral: if the error can take N forms, a model that reproduces training frequencies assigns 0.5 to the correct string and $0.5/N$ to each false one and wins the comparison for every $N \geq 2$, so any experiment that raises N raises pair accuracy for free. We remove this channel by summing over the whole false support,

$$\text{share} \;=\; \frac{P(\text{correct})}{P(\text{correct}) + \sum_i P(\text{false}_i)}.$$

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On the primary denoising corpora each problem occurs exactly twice, once correct and once corrupted, and every other string in the enumeration occurs zero times, so a frequency-matching model scores exactly 0.500 regardless of N.

We train transformers of 3.5M–3.08B parameters from scratch on mathematical corpora where each problem appears with correct and incorrect solutions, vary the structure of the errors at fixed frequency, format and domain, and compute both measures on the same checkpoints. Our contributions are as follows. (a) A measurement of correctness preference that mass splitting cannot inflate, with an exact null, showing 59.0% at 86M against random errors and no more than chance at 3.5M–26M on the primary corpus. (b) Evidence that the coherent false rules of our main corpora produce no preference at any size, immune to splitting because each has one false string, with their apparent sub-chance readings traced to completion length; one single-rule corruption of a different construction does lose mass, so which properties of a rule decide the outcome remains open. (c) Two negative controls, a multi-rule sweep and a description-length ladder, in which pair accuracy (multi-rule) and the realized-support share (ladder) rise by 32 and 16 points while the family share stays flat or falls. (d) A calibration showing that the pairwise measures overstate the excess above chance three- to fourfold in the one condition where all of them can be computed exactly. (e) A selector control that scores its selectors against the same enumerated false family and changes how the training corpus picks its false string; a selector the model can learn moves the family share from $+5$ to -7 points around the null at 418M–3.08B, while the correct completion still outweighs the selected false string. Results elsewhere in the paper that rest on pair accuracy alone are marked as such.

2 Related Work

Prediction as compression. The link between prediction and compression traces to Shannon [1948] and Solomonoff [1964] and to Rissanen's MDL principle [Rissanen, 1978, Grünwald, 2007]. Delétang et al. [2024] showed that language models are strong general-purpose compressors, Huang et al. [2024] found a linear relation between compression quality and benchmark performance, and Wan and Mei [2025] argue that LLM training approximates Solomonoff induction. Pan et al. [2025] link compression to knowledge acquisition, Chlon et al. [2025] link compression failures to hallucination, and contemporaneous preprints formalize simplicity bias through MDL [Marty et al., 2026] and connect compression to mathematical modeling [Aksenov et al., 2026]. This framing motivates our design.

Representations, data statistics and truthfulness. Structured internal representations arise from prediction [Li et al., 2023a, Gurnee and Tegmark, 2024], and several studies locate truth-correlated directions [Burns et al., 2023, Marks and Tegmark, 2024, Li et al., 2023b, Azaria and Mitchell, 2023, Bürger et al., 2024, Ravfogel et al., 2025]; we work at the behavioral level. Truthfulness tracks pretraining "personas" [Joshi et al., 2024] and document frequency [Elazar et al., 2022, Kandpal et al., 2023], and under conflicting knowledge the model follows formality and surface quality [Li et al., 2024], contextual evidence [Longpre et al., 2021] or adaptive conflict resolution [Xie et al., 2024]. TruthfulQA [Lin et al., 2022] shows that models reproduce falsehoods common among humans, consistent with an objective that carries no prior toward truth; our coherent condition is the controlled form of that observation. We fix frequency, format and surface quality and vary only the structure of the error system.

Simplicity bias, noisy labels and poisoning. Neural networks prefer simple functions [Valle-Pérez et al., 2019, Mingard et al., 2021, Goldblum et al., 2024], memorize random labels [Zhang et al., 2017, Arpit et al., 2017] yet tolerate massive label noise [Rolnick et al., 2017], and grokking connects compression to generalization [Nanda et al., 2023, Liu et al., 2023]. The poisoning literature, from BadNets [Gu et al., 2017] through concealed NLP poisoning [Wallace et al., 2021] to web-scale poisoning [Carlini et al., 2024] and near-constant poison counts for LLMs [Souly et al., 2025], studies coordinated, internally consistent corruption, which is the kind of corruption our coherent condition models; in that controlled setting we find no filtering of it by the objective.

3 Methodology

Hypothesis and MDL framing. Consider a corpus in which each problem appears with a correct answer (theory T_1) and an alternative (T_2). An MDL learner [Rissanen, 1978, Grünwald,

2007] minimizing $L(M) + L(D|M)$ must encode *both* clusters as a latent-source mixture, so when $K(T_2) \gg K(T_1)$ (random errors) the false cluster costs a memorized answer per problem, and when $K(T_2) \approx K(T_1)$ (coherent errors) there is no basis to prefer either (Appendix B). This motivated the design, and its graded consequence fails: a wrong rule storing more constants should be filtered more strongly, and Section 4.3 varies exactly that and finds the share flat. We keep MDL as motivation and interpretation, not as a quantity these experiments measure; a gzip proxy for description length (Appendix E) tracks pair accuracy only through the splitting-inflated multi-rule points. Throughout, "truth" means correctness of mathematical derivations and factual accuracy of Wikipedia paragraphs; models compress text, not reality.

3.1 Models and Training

We train decoder-only transformers with a GPT-2-like design [Radford et al., 2019] (pre-norm, GELU, causal mask) at four sizes: *tiny* (4 layers, d_{model}=256, 4 heads, 3.5M parameters), *small* (6×384, 6 heads, 12M), *medium* (8×512, 8 heads, 26M) and *large* (12×768, 12 heads, 86M; Table 6, Appendix A). Section 4.8 and Table 4 extend to the Qwen3 architecture [Yang et al., 2025] at 418M–3.08B. Training uses AdamW (weight decay 0.01), cosine decay with linear warmup, lr $=$ 3e-4, sequence length 256, batch size 32 and 5000 steps; learning curves plateau by step 3000–4000 at all sizes (Appendix H). Core conditions use 4 random seeds, and the seed count is stated with every table. The original denoising experiments use PyTorch [Paszke et al., 2019], the multi-rule and Wikipedia experiments MLX 0.31.0 [Hannun et al., 2023]; the per-experiment hardware and software record is in Appendix A. Problems are generated and verified with SymPy 1.14 [Meurer et al., 2017]; no human annotation is involved, and the generation templates are in the released repository.

3.2 Corpus Design

Denoising setup (primary). A generator writes mathematical problems of four types (multi-step arithmetic, factoring, equation solving, differentiation) as step-by-step derivations in English, and **each problem appears with contradictory answers**: once correct and once with an error. Each corpus contains 5,000 unique problems. The *equal random* condition pairs each correct derivation with one random error (1:1), the *equal coherent* condition with one error produced by a single systematic false rule applied consistently to every problem of a type (1:1), and the 1:2 and 1:4 conditions add two or four random errors per problem (Section 4.6; Table 7, Appendix A). The tokenizer is character-level (vocabulary $\sim$57); BPE robustness is checked in Section 4.6.

Wikipedia setup (transfer). We construct corpora from 20,000 English-Wikipedia paragraphs sampled from the 2023-11-01 Wikimedia dump [Wikimedia Foundation, 2023], with spaCy 3.8 (`en_core_web_sm`) for named-entity recognition [Honnibal et al., 2020], in two corruption modes: *random substitution* replaces each entity with a random entity of the same type, and *coherent substitution* applies one consistent global mapping (every "France" $\rightarrow$ "Japan", every "Paris" $\rightarrow$ "Tokyo"). Both preserve type and grammatical structure; random substitution breaks co-occurrence patterns. Models train on 50/50 mixes and are evaluated on 2,000 held-out paired paragraphs.

3.3 Evaluation

Aggregate share (primary). For each problem we enumerate the full set of completions the error generator can produce, score each with the summed log-probability of its tokens conditioned on the shared prompt (true $\log P(\text{string} \mid \text{prompt})$, not a length-normalized score), and report the share defined in the introduction. The enumeration is exact: each generator has a finite error family, and the enumeration script drives the generator's own rendering functions and verifies for every problem that the error the corpus contains is a member of the enumerated set. Because each problem occurs exactly twice in the primary corpora and every other member of the family occurs zero times, a model that reproduces training frequencies scores exactly 0.500 for any family size. The share is conditional on the enumerated support, not the model's total probability of the correct answer; adding strings to the sum can only lower it, so it is conservative relative to any comparison against a subset of the support, the pairwise one included. The share is also a full-string log-probability conditioned on the shared prompt rather than a per-token mean, so local token predictability inside a completion cannot inflate it through a per-token average, and the length-matched subset below removes the one remaining channel distinct from the preference itself, completion length.

Size	Params	Random errors			Coherent error	
		Aggregate share, % (null 50)		Pair acc., %	Aggregate share, % (null 50)	
		full family	realized	realized	1 mode	equal length
tiny	3.5M	18.6 ± 1.1	65.5 ± 0.9	71.7 ± 1.7	43.5 ± 0.4	46.7 ± 0.5
small	12M	31.6 ± 1.2	76.0 ± 0.9	84.1 ± 1.0	45.8 ± 1.1	48.3 ± 0.8
medium	26M	44.1 ± 0.7	80.3 ± 0.8	89.2 ± 1.4	46.4 ± 1.5	48.1 ± 0.9
large	86M	$\mathbf{59.0 \pm 0.8}$	77.6 ± 0.2	87.4 ± 0.4	48.8 ± 0.4	49.3 ± 0.4

Table 1: Denoising corpus, measured on training problems where the null is exact (4 seeds, mean $\pm$ SD, %). The columns differ only in how much of the false side is summed: the generator's full support (a mean of 12.0 false modes per problem, range 4–35; n=2569), the single error the corpus contains (n=2620), or, for pair accuracy, that same single error under mean NLL. The coherent condition has one false mode, so full and realized coincide (n=2437).

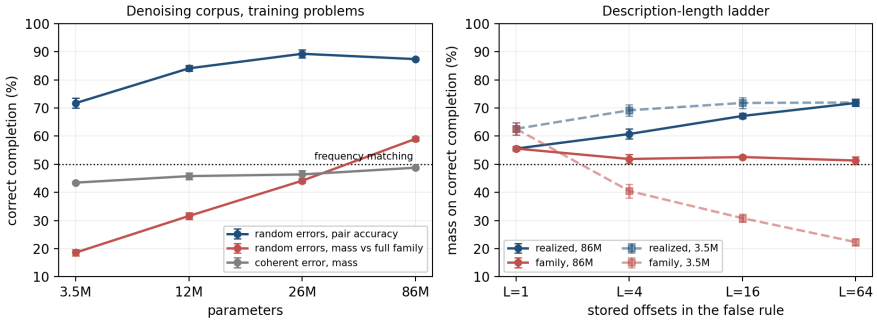


Figure 1: Left: the denoising grid under both measures (4 seeds). Comparing pairwise against the single realized error reports a large effect at every size; summing the generator's full support leaves an effect only at 86M, and the coherent condition sits just below the frequency-matching line throughout (a completion-length effect; the length-matched values approach chance). Right: the description-length ladder of Section 4.3, where summing only the realized false string makes the share rise with the number of offsets the rule stores and summing the full family does not.

Statistics. Per-run 95% confidence intervals are bootstrap intervals over problems (5000 resamples); the $\pm$ attached to a mean across seeds is the population standard deviation across seeds; pair-accuracy evaluations also store a one-sided Wilcoxon signed-rank p on per-pair NLL differences and a bootstrap interval, released with the per-run outputs.

Pair accuracy (secondary). Pair accuracy is the fraction of pairs where the model assigns lower mean NLL to the correct completion than to one sampled incorrect completion, equivalent to the Common Language Effect Size [McGraw and Wong, 1992]. Reported for comparability with the prior literature, it is not a controlled diagnostic of preference: whenever the false side has more than one form, a frequency-matching model wins it without preferring anything (Section 4.1).

Length control. Every measurement records the mean difference in completion length between the false and correct sides and the share restricted to problems where all completions have identical token length. Shorter strings carry more probability by construction (Section 4.1).

Generative evaluation. We decode greedily on 500 test prompts and verify each answer with SymPy (Section 4.7).

4 Results

4.1 The Central Contrast: Random vs Coherent Errors

The aggregate share puts the correct completion at $59.0\% \pm 0.8\%$ of the probability mass at 86M parameters against everything the error generator can produce (Table 1), with all four per-seed bootstrap intervals above 0.5: $[58.7, 60.4]$, $[57.5, 59.3]$, $[57.2, 59.0]$, $[59.0, 60.7]$. Because adding

N rules	Distinct false modes	Pair accuracy, %	Aggregate share, % (null 50)
1 (coherent)	1.00	47.0 ± 2.2	47.9 ± 0.7
2	2.00	79.1 ± 1.1	47.8 ± 0.4
3	2.91	83.1 ± 0.4	41.8 ± 0.6
5	4.26	86.1 ± 0.5	41.9 ± 0.5
10	8.52	90.1 ± 0.3	38.7 ± 1.0

Table 2: Multi-rule errors (tiny, 3.5M, standard 50/50, 4 seeds, %). Pair accuracy rises by 43 points across the sweep; the aggregate share, measured on the same checkpoints against the enumerated false support, does not rise at all. The frequency-matching null is 50.0 for the share column at every N. The N=1 row is scored on the multi-rule family's own N=1 support (n=916); the coherent entry of Table 10 (47.3) uses the coherent support (n=915).

strings to the denominator can only lower the value, 59.0 is conservative relative to the realized-error share. An independent corpus gives a matching value at that size: on the standard (non-denoising) corpus, with a different generator and a mean of 5.51 false modes per problem, the share is $59.4\% \pm 0.6\%$ (Appendix F).

Below 86M the aggregate share shows no preference: 18.6%, 31.6% and 44.1% at 3.5M, 12M and 26M, all below the null, while pair accuracy at those sizes already reads 72–89%. Part of that shortfall is a downward bias of the measure at small scale rather than a preference for the false system. For models trained on the ladder corpus of Section 4.3 with no false rule anywhere, the family share on its 18-string arithmetic support reads 99.7% at 86M but only 64.1% at 3.5M: a third of the mass at the smallest size leaks onto strings that never occurred. We have no clean calibration on the denoising support itself and take the ladder figure as the best available estimate of the leak.

The gap between the measures is what the pairwise comparison adds on its own. At 86M, scoring the correct completion against the single realized error gives 27.6 pp of excess above chance where summing the family gives 9.0 pp, a threefold overstatement, and pair accuracy against that same error gives 37.4 pp, a fourfold one. Below 86M the measures disagree about whether there is an effect at all.

Against a coherent false rule no preference appears at any size: 43.5, 45.8, 46.4 and 48.8%. One rule produces one string, so there is nothing to split, and this half of the contrast is immune to the objection by construction. The readings sit slightly below 0.5 because the coherent false completions are on average 0.58 tokens shorter than the correct ones; on the length-matched subset the values are 46.7, 48.3, 48.1 and 49.3%, approaching chance from below rather than sitting under it. An earlier draft attributed the sub-chance readings to the textual simplicity of the false rules; that reading was wrong, and we draw no conclusion from the shortfall.

4.2 Multi-Rule Errors: A Negative Control for Mass Splitting

The first version of this work made this experiment its central identification argument: models train on corpora with N compact wrong rules per task type, each problem's rule drawn at random, so that only the predictability of the problem-to-rule mapping changes, and pair accuracy jumps from 47% at N=1 to 79% at N=2. That jump is mass splitting. Aggregate share is $47.9\% \pm 0.7\%$ at N=1 and $47.8\% \pm 0.4\%$ at N=2 (Table 2; Figure 2, Appendix C): the transition that carries 32 points of pair accuracy moves the allocation between the two answer systems by 0.1 points. We had argued that splitting could not explain the jump because complete splitting would drive N=2 accuracy toward 100%; partial separation predicts what is observed, and the direct measurement supports that reading. The experiment is withdrawn as evidence for a selector cost and reported as a calibration of how much pair-accuracy signal splitting alone can manufacture. Beyond N=2 the decline is partly a property of the measure, which collects more diffuse uncertainty as more neighboring strings are summed; the safe reading is the absence of an increase.

4.3 A Description-Length Ladder, and Why It Fails

The multi-rule design confounds description length with mode count. A cleaner test varies the description length of a *single deterministic* false rule, $a \cdot b + T[\,|a| \bmod L\,]$ with T a fixed table of L offsets, so that L is the number of constants the rule stores while the computation, the corruption

Aggregate share, % (86M, null 50)	L=1	L=4	L=16	L=64
Realized, 1 string	55.6 ± 0.7	60.7 ± 1.8	67.2 ± 0.9	71.8 ± 1.2
Full family, 1–18 strings	55.6 ± 0.7	51.9 ± 1.6	52.6 ± 0.6	51.3 ± 1.3

Table 3: Description-length ladder at 86M (4 seeds, n=1339–1342 problems, %). *Realized* scores the correct completion against the single false string the corpus contains; *family* scores it against every string the rule could have emitted for that problem. Those other strings occur zero times in the corpus, so both measures have a 50.0 null at every rung, and the two coincide at L=1 by construction. All four sizes are in Table 8, Appendix C.

load, the error magnitudes and the task distribution stay fixed and completion lengths differ by at most 0.044 tokens. If preference were graded in description length, it should rise with L. Over the realized support it does, at every size. Summing the full family removes the rise and reverses it below 86M (Table 3, Figure 1 right); at 86M the family reads 55.6, 51.9, 52.6, 51.3. The realized rise has an explanation that owes nothing to description length: a model that has not learned the table spreads its false mass over the entries and is credited for the spread.

Calibration. The same four sizes trained on a corpus with the same problems and *no false rule anywhere* (2 seeds) give an 18-string family share of 99.7% at 86M, 97.3% at 26M, 93.2% at 12M and 64.1% at 3.5M (Table 9, Appendix C). At the top size a clean model reads close to 100, which bounds the leak only for a clean model; at the bottom the leak of Section 4.1 makes small-model family numbers conservative rather than evidence about preference. With Section 4.2, both experiments built to identify a compression mechanism measure something else. The L=1 rung is itself a single coherent rule and sits above the null at every size (55.6 ± 0.7 at 86M, 62.6 ± 2.2 at 3.5M; every per-seed interval above 0.5, one false string, lengths matched), so the absence of filtering for coherent rules is a property of the coherent conditions of Sections 4.1, 4.4, 4.5, 4.6 and 4.8, not of every single-rule corruption. The ladder rule corrupts only the first multiplication and leaves the rest of the derivation correct, and we have no measured account beyond that.

4.4 The Learnable-Selector Control

Both withdrawn designs move the metric by changing how many false strings a problem has. This control holds the scored family fixed: every selector is evaluated against the same enumerated random family ($\approx$17.5 strings per problem) and the conditions differ in how the training corpus picks the one false string it writes. In `learnable` that string is $a{\cdot}b + T[\,s(p) \bmod 10\,]$ for a fixed 10-entry offset table T, reaching about ten family members per problem, with the selector $s(p)$ one visible operand's ones digit, a lookup the model can fit. In `k2` and `k3` it is the sum of two or three ones digits mod 10, learnable in principle and progressively harder to fit. In `learnable_mod` the index is the sum of all visible operands modulo the problem's family size, bypassing T; it is deterministic but not learned at any size. These sit alongside `random` (a uniform draw from the family) and `coherent` (one rule), family shares scored on the length-matched subset of the held-out scale grid (500k texts, 0.44 epochs, evaluation problems disjoint from training by a hash, null 0.500 up to a $\leq$1.3-point correction; Table 11, Appendix D), with a clean-corpus calibration at every size; realized shares at every size are in Table 12.

With the scored family held fixed, the family share's position relative to the null tracks learnability (Table 4). Against a selector the model cannot learn (`random`, `learnable_mod`) the correct completion holds two to seven points above the null at every size; against one it can (`learnable`) the share falls below the null, to $43.4\% \pm 0.6\%$ at 474M (4 seeds). The selected false string does not outrank the correct one: against it alone the correct completion holds 65.1% at 474M (64.2% length-matched; Table 12), and the below-null family share comes from the mass on the other members of the family. The intermediate selectors sit between the two ends: as s depends on more digits the model fits it less, the realized false string carries less mass ($65 \to 89 \to 92$ at 474M) and the family share rises from below the null to the null ($43 \to 49 \to 49$ at 474M; $45 \to 49 \to 48$ at 418M); the unlearnable selectors sit above it.

A below-null family share does not by itself show a shift toward a false system: it can also come from lower confidence in the correct completion, as in a model that has not learned the arithmetic (Appendix D), and the saved outputs (mass ratios; weights deleted) cannot separate the two. Com-

Family share, % (null 50)	418M	474M	1.12B	2.16B	3.08B	Realized share, 474M
`random` (unpredictable)	56.8 ± 1.1	54.9 ± 0.2	55.5 ± 0.1	54.5 ± 0.2	56.6 ± 0.5	—
`coherent` (one rule)	50.2 ± 0.4	49.9 ± 0.5	49.5 ± 0.1	50.7 ± 0.3	49.3 ± 0.3	—
`learnable` (1 digit)	44.7 ± 1.0	$\mathbf{43.4 \pm 0.6}$	43.8 ± 0.1	43.1 ± 0.7	43.7 ± 1.2	65.1 ± 1.6
`k2` (2 digits)	48.6 ± 0.0	49.2 ± 0.1	—	—	—	88.6 ± 0.1
`k3` (3 digits)	47.6 ± 1.4	48.8 ± 0.1	—	—	—	91.6 ± 0.2
`learnable_mod` (all operands)	56.1 ± 1.1	54.8 ± 0.6	—	53.2 (1 seed)	54.6 ± 0.0	92.9 ± 0.2
clean (no false rule)	97.9 ± 0.3	97.2 ± 0.1	98.1 ± 1.2	96.4 ± 0.2	99.0 ± 0.3	—

Table 4: The learnable-selector control and its replication to 3.08B (Qwen3 architecture; family shares length-matched, realized full-set). The frequency-matching null is 50, or at most 51.3 after the marginal correction of Appendix D. `random`, `coherent` and `learnable` are 4 seeds at 418M and 474M; every other cell is 2 seeds except `learnable_mod` at 2.16B (1 seed; not run at 1.12B). Rows other than `coherent` are scored against the same random family; `learnable`, `k2` and `k3` write from one 10-entry table and differ in how learnable its index is. *Realized* scores the correct completion against the single false string the corpus used (`random` has none on held-out problems); *clean* trains on the same generator with no false rule and calibrates the leak.

petence is not matched by construction either: the random and learnable corpora share a generator and a problem distribution but not their problems, because the random draw consumes the shared random stream. What the control does show is that predictability, not description length, moves the family share: the ladder varied the number of constants a deterministic rule stores, with the table increasingly unlearned at higher L (Section 4.3), and the family share did not rise; here the stored table is fixed, the learnability of its index varies, and the share crosses the null. One caution remains: `learnable_mod` writes from the full family rather than the 10-entry table, so it anchors the unlearnable end as a direction rather than a matched-diversity point; the clean matched-diversity gradient is `learnable`→`k2`→`k3`. The contrast holds at 1.12B, 2.16B and 3.08B parameters (2 seeds per cell): `random` above the null, `coherent` at it and `learnable` below it, with the clean calibration above 96%, so the measure's leak on a clean model is small at these sizes.

4.5 Transfer to Wikipedia Text with Entity Substitution

In pair accuracy the two conditions again separate (Table 18, Appendix G): 69.5–71.5% under random substitution and 45.9–48.7% under coherent substitution across 3.5M–86M (4 seeds). Only the coherent column is interpretable: one global mapping yields one false paragraph per prompt, its null is 50%, and a consistent substitution is not filtered on real text either. Random substitution replaces every entity site (a median of eight per paragraph), its support cannot be enumerated, and on held-out paragraphs a frequency-matching model wins nearly every comparison, so that column's null is close to 100% and we draw no conclusion from it or from its per-entity-type breakdown (Table 20). The false paragraph is longer by 28.1 characters under random and 27.8 under coherent substitution, so length does not explain the difference between the conditions, but only 1% of pairs are length-matched and neither column has the length control of Section 4.1.

The enumerable special case, single-entity paragraphs, quantifies why natural text resists the measure (Table 19, Appendix G). Its random support is one entity type's pool of ∼2800 strings: a clean 86M model with no false rule reads 5.6% of the mass on the correct string, the random-trained model 4.0%, both leak-dominated. On the one-mode coherent support the length-matched subset, pooled over 8 runs (n=25 per cell), reads 52.4 ± 4.0 at the 50 null, consistent with the paired result but underpowered per cell.

4.6 Robustness Checks

Standard corpus. In the standard setup each problem appears once, correct or incorrect, and problems repeat often enough that frequencies are exactly known. The aggregate share reads 50.2 ± 0.7, 57.6 ± 0.2, 59.0 ± 0.3 and 59.4 ± 0.6 for random errors and 47.3 ± 0.7, 49.2 ± 0.4, 50.0 ± 0.4 and 50.1 ± 0.1 for coherent ones (4 seeds; 5.51 false modes on average). Here the contrast appears from 12M, two sizes earlier than on the denoising corpus, and is absent at 3.5M, where pair accuracy on the same support nevertheless reads $76.3\% \pm 1.5\%$ (Table 10; the standard-versus-denoising pair-accuracy comparison is Table 14).

Size	Random-trained	Coherent-trained	Gap
	generative accuracy, %; greedy decode, SymPy-verified		pp
tiny (3.5M)	$31.1\% \pm 1.4\%$	$20.8\% \pm 3.6\%$	$+10.3$ pp
small (12M)	$46.7\% \pm 1.8\%$	$31.7\% \pm 1.7\%$	$+15.0$ pp
medium (26M)	$50.5\% \pm 1.3\%$	$36.1\% \pm 2.6\%$	$+14.4$ pp
large (86M)	$52.8\% \pm 1.0\%$	$35.5\% \pm 1.8\%$	$+17.3$ pp

Table 5: Generative accuracy, greedy decoding with SymPy verification (4 seeds, 500 problems each). No evaluator-supplied completion is scored, so no frequency-matching null applies; the argmax is still sensitive to how the false mass is divided. Every cell is recomputed from the released per-run evaluation logs; the tiny random-trained cell was re-run in August 2026 on the surviving checkpoints after the original logs were lost.

Noise ratio on the controlled measure. With K random errors per problem the frequency-matching null is exactly $1/(K{+}1)$. At 86M (2 seeds, non-strict support of 4565 problems with a mean of 8.2 false modes) the family share reads 56.3 ± 0.7, 44.9 ± 0.2 and 36.1 ± 0.7 at $K{=}1, 2, 4$ against nulls of 50.0, 33.3 and 20.0. The absolute share falls as pair accuracy on the same support does ($88.7 \rightarrow 79.4 \rightarrow 76.3$; the original pair-accuracy-only sweep is Table 16), but the excess above the null does not shrink: $+6.3$, $+11.6$, $+16.1$ on the full set and $+7.3$, $+7.6$, $+11.4$ on the length-matched subset (Table 17, Appendix F). Its $K{=}1$ value is below the strict 59.0 because the non-strict support keeps problems whose strings collide across problems. We read this as the preference not shrinking with noise, not as strengthening.

Tokenization. Under a BPE tokenizer (SentencePiece, vocabulary 1000) the coherent condition stays at chance in pair accuracy, and its denoising value moves from 43.5% at character level to 49.3%, consistent with the length account; the random column rises too, but on a multi-mode support that rise is not interpretable (pair accuracy only; Table 21).

Frequency. For coherent errors frequency dominates: at 40/60 the model prefers the false system in 72% of pairs and at 20/80 in 90% (Table 15), numbers that one false string leaves unaffected by splitting. For random errors the minority regime has only pair accuracy, and the aggregate share at 50/50 in that setup is already at chance (50.2 ± 0.7), so we claim no persistence of a random-error preference there.

Matched control. "Matched random" errors that corrupt as many steps as the coherent rule give $74.5\% \pm 0.4\%$ pair accuracy against 47.2% for coherent, which rules out corruption load but not mode count (Appendix F); a chained-verification variant that raises coherent pair accuracy is reported, unmeasured on the share, in Appendix H.2.

4.7 Generative Evaluation

Greedy decoding scores no false completion, but it is not immune to splitting: when the false mass is divided among many alternatives, the argmax can land on the correct answer without any gain in its mass. Below 86M it is the only measurement that separates the conditions at every size; the aggregate share does so from 12M on the standard corpus and not at all on the denoising corpus (Section 4.6). Random-trained models solve more problems than coherent-trained ones at every size, and the gap is descriptively larger at 86M than at 3.5M ($10.3 \rightarrow 17.3$ pp, Table 5). At 12M, where the controlled likelihood measure reads 31.6% and reports no preference, generation separates the conditions with no per-seed overlap, $46.7\% \pm 1.8\%$ against $31.7\% \pm 1.7\%$. Generation records the argmax, not the share of mass, so the gap below 86M is a behavioral difference, not evidence of a preference. Multi-step arithmetic is never generated correctly at these sizes (per-task breakdown in Appendix H.3).

4.8 Architecture and Scale: Qwen3 at 418M and 490M

Two checks address whether the contrast is tied to the GPT-2-like design or to toy scale. A Qwen3-0.6B configuration [Yang et al., 2025] (RoPE, GQA, SwiGLU, RMSNorm; 417.6M parameters with our character vocabulary, labeled 418M throughout) trained from scratch on 50/50 corpora of the standard construction scaled to 1.4M problems (35k steps, batch 16, sequence length 256, bf16; 3 seeds) gives $87.3\% \pm 0.4\%$ pair accuracy under random errors and $49.4\% \pm 1.3\%$ under coherent

ones. On the same checkpoints the aggregate share puts the correct completion at $56.3\% \pm 0.3\%$ against the enumerated support for random errors and $49.7\% \pm 0.0\%$ for coherent ones (null 50%). The contrast holds on the controlled measure at 418M as at 86M, and a coherent rule is not filtered on either metric.

A Qwen3-architecture model of 490M parameters (labeled $\sim$1B in an earlier draft; it instantiates to 489.9M at the run's BPE-8000 vocabulary) was trained from scratch on 1 GB of FineWeb-Edu [Penedo et al., 2024] with mathematical content filtered out and replaced by 8% contradictory math (15,000 steps, batch 4, sequence length 512, lr 3e-4 with 200 warmup steps, fp16, $\approx$31M tokens). It gives $77.6\% \pm 0.8\%$ pair accuracy under random errors and $47.4\% \pm 0.7\%$ under coherent ones (2 seeds, the first on a Kaggle T4 and the second on an H100). This is pair accuracy on an enumeration-free support, so the random value carries no quantitative claim; what the run establishes is that a coherent false rule is not filtered at 490M on real text with the contradictory material in the minority. The replication of the controlled contrast at 1.12B, 2.16B and 3.08B is in Table 4.

5 Discussion

Summary of findings. Two math corpora agree at 86M: a false system whose output the model cannot predict from the problem loses about nine points of probability mass to the correct one (59.0% denoising, 59.4% standard), and a coherent false rule loses none in any coherent condition we built, on every corpus and at every size we tested (the one exception is the ladder's first rung, a single-rule corruption of a different construction; Section 4.3). Below 86M the corpora disagree, so the effect is established at one size and nowhere below, while greedy generation, a behavioral measure that splitting can also move, separates the conditions at every size. Section 4.4 points to the axis that moves the share: against the same scored family, a selector the model can learn moves the correct share from $+5$ to -7 points around the null, and the partly learnable selectors that share its table sit at the null; whether the below-null cells reflect a shift toward the false system or lower confidence in the correct one is not resolved.

What we withdraw. Preference is not graded in the description length of the false system. The two experiments built to show it move the pairwise readings by 32 and 16 points (pair accuracy in one, the realized share in the other) while leaving the family share flat or falling; their apparent signal is the model's inability to predict which false string will appear, credited to the correct side by a pairwise comparison. Coherent errors do not fall below chance (that was completion length), and the effect is not present at every scale. None of these three claims from the first version survives its own control. Pair accuracy remains a legitimate measure of ranking behavior, but any design that varies the number of false forms (competing rules, noise ratios, entity substitution, corruption load) varies the metric along with the treatment.

The role of frequency. Coherent falsehood needs no frequency advantage to neutralize any preference for the correct answer (Section 4.6); for random errors the excess above the frequency-matching null persists to four errors per problem at 86M, and on the standard corpus, where only pair accuracy exists, we make no minority-regime claim.

Implications and scope. Within these settings the training objective offers no inherent "truth compass". In the coherent conditions we built, internally coherent falsehood stays competitive at every size, on synthetic math and on diluted natural text (the ladder's first rung is the exception); the preference against incoherent falsehood is real but modest where we can measure it cleanly; and a false selector the model *can* learn pushes the correct share below the null (Section 4.4). What goes unfiltered is consistent or learnable structure, not falsehood as such. That bears on data curation, where coordinated errors would evade diversity-based filtering, and on the reading of hallucination as a corpus-statistics phenomenon [Kalai and Vempala, 2024]. Our models remain far below production scale, compression remains an interpretation rather than a measured quantity, and we make no claims about production models.

6 Conclusion

Measured against everything an error generator can produce rather than against one sampled error, correctness preference in contradictory training data is smaller than the pairwise metric reports and appears later: at 86M the correct completion holds 59.0% of the probability mass where the pairwise

comparison reports 77.6%, and below 86M the aggregate share shows no preference on the denoising corpus while pair accuracy reads 72–89%. A single coherent false rule is not filtered in any of our coherent conditions, at any size, architecture or corpus we built them on; the one single-rule corruption outside those conditions, the ladder's first rung, sits above the null, so which properties of a rule decide the outcome is open. The share does not scale with the description length of the false system; it moves with predictability: making the selector of an enumerated false family learnable turns the correct system's small advantage into a share below the null, from 418M to 3.08B parameters, although the correct completion still outweighs the selected false string. The behavioral finding is that what these models leave unfiltered is consistent or learnable false structure, not falsehood as such.

Limitations

Measurement. The aggregate share is exact only where the false support can be enumerated, which rules out natural-text settings in general, and it is a share conditional on that support, not the model's total probability of the correct answer. It carries a downward bias that grows as the model shrinks (clean-corpus family share 99.7% at 86M, 64.1% at 3.5M; Section 4.3) and that depends on architecture at fixed size, a 413M GPT-2 leaking like a 202M one (Appendix D). Where we report an effect, a clean model reads near 100, which bounds the leak but does not by itself exclude a bias in models trained on contradictions; where we report none, a small preference could be hidden by the leak.

Scale, in both directions. Under the aggregate share the effect is present only at the largest primary size on the denoising corpus and from 12M on the standard corpus, so there is one size at which it is established on both; on the held-out grid the random cells clear the null only from 418M, because the leak dominates below. The billion-scale replication (1.12B–3.08B, 2 seeds, 0.44 epochs) exists; what remains open is compute-matched training at that scale and controlled evidence below 86M, where only generation separates the conditions at every size. A same-size 413M GPT-2 reproduces the above-null random cell only marginally (51.2 ± 1.1 against a clean calibration of 93.8), so an architecture component in the effect cannot be excluded (Appendix D).

Domain. Wikipedia entity substitution has no enumerable false support in general, and the single-entity special case (Section 4.5, Table 19) quantifies why rather than removing the obstacle: a $\sim$2800-string random support leaks 94–98 points at 3.5M–86M, and the coherent support's length-matched subset holds only $n{=}25$ problems per cell. What transfers cleanly is the coherent result. Domains with competing real-world knowledge systems remain open.

Discriminative versus generative. Pair accuracy (87.4%) exceeds generative accuracy (52.8%) at 86M, the familiar teacher-forcing gap, whose roots in next-token training are analyzed by Bachmann and Nagarajan [2024]. The generative result stands on its own.

Causal identification. The learnable-selector control changes how the corpus picks its false string while the scored family stays fixed, and its limits are substantial. The random and learnable corpora share a generator but not their problems, so competence is not matched. The below-null share can come from a shift toward the false system or from lower confidence in the correct completion; the saved ratios cannot separate these, and against the selected string alone the correct completion still wins. A learnable selector is both more predictable and cheaper to store, so a preference for the *more compressible* system is consistent with the swing but not measured by it. A rerun with problems fixed in advance, an independent random stream for the false draws and absolute log-probabilities would address the first two. All of it is one formal-math corpus family.

Ethical considerations. Internally consistent misinformation can be harder for models to filter than diverse errors; this is a controlled diagnostic finding, not evidence of production-LLM vulnerability, and transparent reporting of such failure modes can inform defensive data curation. Wikipedia text is used under CC BY-SA 4.0 and FineWeb-Edu under ODC-By 1.0.

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A Reproducibility

The public repository `https://github.com/Rai220/compression-drives-truth` contains all code, data generators, support-enumeration scripts, per-run metric JSON for every cluster grid and SHA-256 checksums of the regenerable corpora. More than 500 training runs were made across all conditions: 218 on a local Apple M4 Pro (48 GB) with MLX 0.31.0 (standard math, multi-rule, Wikipedia, generative and the original robustness conditions; weights retained) and 335 on clusters (weights deleted after evaluation, metric JSON retained). The original denoising runs used Modal.com T4 GPUs with PyTorch 2.3; the denoising and ladder control grids of Sections 4.1 and 4.3 and the held-out scale grid to 474M ran on 8×H100 nodes (per-run timing and PyTorch version not recorded). For the Qwen3-0.6B 50/50 standard-construction corpus of Section 4.8 the numbers come from a 3-seed rerun on a single H100 (the artifacts of the original Modal A10G run were never retrieved); the 490M FineWeb-Edu run has seed 42 on a Kaggle T4 and seed 43 on the single H100. That single H100 (PyTorch 2.13, bf16 with gradient checkpointing at $\geq$1B, fp32 below) also ran the 1.12B–3.08B grid, the `learnable/k2/k3/learnable_mod` conditions, the clean controls at the Qwen sizes, the Wikipedia single-entity calibration and the K-ratio sweep, about 45 GPU-hours over 81 logged runs; a 4.39B configuration exceeds one H100's memory with fp32 optimizer states and was not run.

Released with this paper. The corpora regenerate byte-identically from their seeds and we publish their checksums; the enumeration script verifies, for every problem, that the error the corpus contains is a member of the enumerated support, and aborts otherwise. Per-run outputs are released for both control grids (64 ladder runs plus 8 clean controls; 32 denoising runs), for the held-out scale grid including its clean, learnable and billion-scale cells, for the K-ratio sweep, the Qwen3-0.6B 50/50 rerun, the FineWeb-Edu runs and the Wikipedia single-entity evaluations.

Corpus generation example. Each problem is generated programmatically and verified by SymPy. A representative denoising training example (equal random condition):

```
Problem: Expand and simplify 3 * (2x + 5) - 4x
Step 1: 3 * (2x + 5) = 6x + 15
Step 2: 6x + 15 - 4x = 2x + 15
Answer: 2x + 15
```

Config	Layers	d_{model}	Heads	Parameters
tiny	4	256	4	$\sim$3.5M
small	6	384	6	$\sim$12M
medium	8	512	8	$\sim$26M
large	12	768	12	$\sim$86M

Table 6: GPT-2-like model configurations used in the primary experiments (Section 3.1).

Condition	Correct	Incorrect	Ratio	Purpose
Equal random	1	1 random	1:1	Primary: preference against unpredictable errors
Equal coherent	1	1 coherent	1:1	Control: one consistent false rule, one false string
2:1 noise	1	2 random	1:2	Noise ratio (null $1/3$ on the family share)
4:1 noise	1	4 random	1:4	Noise ratio (null $1/5$ on the family share)

Table 7: Denoising corpus conditions (Section 3.2); each corpus contains 5,000 unique problems.

The random-error version replaces a derivation step with a plausible but incorrect computation (e.g., `6x + 15 - 4x = 2x + 11`). In the coherent condition a systematic rule is applied consistently across all problems of the same type (e.g., for distribution, $a(b+c) = ab + c$ instead of $ab + ac$). The paired test set presents the same prompt with both completions for NLL comparison.

B The MDL Framing, and What It Predicted

This appendix records the reasoning that motivated the experimental design, and what happened to it. All of it describes an idealized two-part MDL learner [Rissanen, 1978, Grünwald, 2007], not a theorem about gradient-descent training.

Setup. A corpus D of n problems, each appearing with an answer from a correct rule system T_1 and one from an alternative T_2. The model must encode *both* clusters, which we model as a latent-source mixture with per-problem assignments $z_i \in \{1, 2\}$:

$$L(M) + L(D|M) = \underbrace{L(T_1) + L(T_2)}_{\text{theories}} + \underbrace{L(\{z_i\})}_{\text{source assignments}} + \underbrace{\sum_i L(a_i|T_{z_i})}_{\text{answers given sources}}$$

A learner that internalizes the more compressible theory encodes its answers cheaply and pays a higher per-token cost for the other cluster.

Prediction 1 (random errors): if T_1 is compact and T_2 is a set of independent random answers, the correct cluster holds more probability mass. *Partly borne out.* That quantity is the aggregate share of Section 3.3. It exceeds 0.5 at 86M ($59.0\% \pm 0.8\%$) and fails at 3.5M–26M on the denoising corpus (18.6%, 31.6%, 44.1%), while the standard corpus shows it from 12M (Table 10).

Prediction 2 (coherent errors): if $L(T_1) \approx L(T_2)$, then at equal frequency the two clusters receive comparable mass. *Borne out* in the coherent conditions we built, at every size, architecture and corpus we tested (Sections 4.1, 4.4, 4.5, 4.6, 4.8); the single-rule $L{=}1$ rung of the ladder, which sits above the null at every size, is the one measured exception (Section 4.3).

Prediction 3 (multi-rule errors): N compact rules with random assignment to problems cost an extra $O(n \log N)$ bits for the selector, which should restore the correct cluster's advantage. *Not borne out.* Mass on the correct cluster reads 47.9, 47.8, 41.8, 41.9 and 38.7% at $N = 1, 2, 3, 5, 10$ (Section 4.2), and the ladder of Section 4.3 is flat at 86M (55.6, 51.9, 52.6, 51.3).

The graded reading of this framework is not supported by our own measurement, and only its binary consequence (Prediction 2, and Prediction 1 at our largest size) survives. One prediction the graded reading did not make, but the learnability account does, fits the data: the extra cost the mixture pays for the false cluster is the source-assignment term $L(\{z_i\})$, paid only when z_i cannot be recovered

from the problem. Making it recoverable, as the learnable selector of Section 4.4 does, should remove the cost and let the model commit to the false cluster. The family share does drop below the null there while the ladder, which left the assignment unpredictable at every rung, never moved; this is consistent with the reading but does not establish it, since the correct completion still outweighs the selected false string and the below-null share may reflect lower confidence in the correct one (Section 4.4). We report compression as an interpretation of the surviving contrast, not as a quantity these experiments measure.

C Control Grids in Full

C.1 Description-Length Ladder, All Sizes

Params	Support (share, %, null 50)	L=1	L=4	L=16	L=64
3.5M	realized	62.6 ± 2.2	69.2 ± 2.0	71.8 ± 2.0	71.9 ± 1.2
	family	62.6 ± 2.2	40.5 ± 2.5	30.8 ± 1.4	22.3 ± 1.2
12M	realized	63.0 ± 1.4	72.6 ± 1.4	80.7 ± 0.9	81.8 ± 1.2
	family	63.0 ± 1.4	49.6 ± 1.9	45.1 ± 1.8	36.2 ± 1.8
26M	realized	57.5 ± 1.2	68.0 ± 0.9	77.4 ± 1.6	81.6 ± 1.4
	family	57.5 ± 1.2	50.8 ± 1.6	51.8 ± 1.8	43.9 ± 1.9
86M	realized	55.6 ± 0.7	60.7 ± 1.8	67.2 ± 0.9	71.8 ± 1.2
	family	55.6 ± 0.7	51.9 ± 1.6	52.6 ± 0.6	51.3 ± 1.3

Table 8: Description-length ladder, 4 rungs $\times$ 4 sizes $\times$ 4 seeds (%). Completion lengths differ by at most 0.044 tokens across all 64 runs, and the length-matched subset reproduces these values to within 1.1 points.

Support summed (share, %; ideal 100)	3.5M	12M	26M	86M
Full family, 18 strings	64.1 ± 4.4	93.2 ± 0.6	97.3 ± 1.3	99.7 ± 0.0
Realized, 1 string	95.1 ± 1.0	99.6 ± 0.1	99.9 ± 0.0	100.0 ± 0.0

Table 9: Calibration control: the same measures applied to models trained on a corpus containing the same problems and *no false rule anywhere* (2 seeds, %). The true value is 100 by construction, so the shortfall is the measure's downward bias: 0.3 points at 86M and 36 points at 3.5M. The ± 4.4 at 3.5M is a spread of 59.7 to 68.5 across two seeds.

C.2 Multi-Rule Sweep

Figure 2 plots the pair-accuracy and aggregate-share columns of Table 2 (Section 4.2) against the number of rules.

C.3 Standard Corpus, Aggregate Share

Aggregate share, % (null 50)	3.5M	12M	26M	86M
Random (5.51 modes)	50.2 ± 0.7	57.6 ± 0.2	59.0 ± 0.3	$\mathbf{59.4 \pm 0.6}$
equal length	53.2 ± 1.1	57.9 ± 0.4	57.8 ± 0.2	58.5 ± 0.7
Coherent (1 mode)	47.3 ± 0.7	49.2 ± 0.4	50.0 ± 0.4	50.1 ± 0.1
equal length	49.3 ± 0.7	49.5 ± 0.4	49.9 ± 0.4	49.7 ± 0.2

Table 10: Aggregate share on the standard (non-denoising) corpus, 4 seeds (%); n=1101 problems for random and 915 for coherent, of which 359 and 561 are length-matched. The 59.4 at 86M is the independent replication of the 59.0 obtained on the denoising corpus. Per-seed bootstrap intervals for the random condition lie entirely above 0.5 at 12M, 26M and 86M. The coherent intervals contain 0.5 from 26M upward; at 3.5M all four lie entirely below it, and at 12M one of four does, consistent with the length account of Section 4.1.

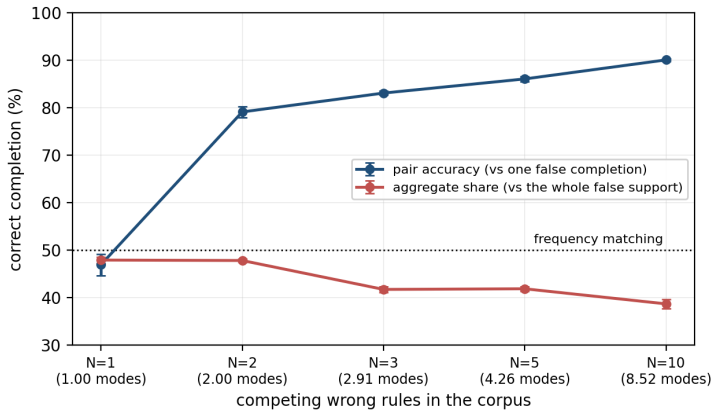


Figure 2: Multi-rule errors under both measures (tiny, 4 seeds). Pair accuracy rises monotonically with the number of distinct false modes while the model's allocation between the correct and false clusters does not move at $N{=}1 \to 2$ and declines thereafter. The x-axis is categorical.

D The Held-Out Scale Grid and the Learnable-Selector Control

This grid trains one model per condition on 500k texts for 5000 steps at batch 32 (0.44 epochs; training windows are sampled with replacement, so some text recurs) and evaluates on $\approx$2.5k held-out problems disjoint from training by a hash of the problem parameters. It is the measurement behind Table 4. Because each text is correct with probability 0.5 and otherwise carries one false string, the frequency-matching null for the family share is exactly 0.500 (a marginal correction from arithmetic problems that render identically in every mode raises it to at most 0.513; we read the band $[0.500, 0.513]$, i.e. 50.0–51.3%). We report the length-matched subset, on which the completion-length confound of Section 4.1 is removed.

A null of indifference is also a null of competence. Every null here describes a model that both lacks a preference *and* can compute the correct answer; a model that prefers truth but has not learned the arithmetic reads below it too. So a below-null cell is not on its own evidence of a preference for the false side. Only excursions above the band are unambiguous. This applies to the learnable control too: its random and learnable corpora share a generator and a problem distribution but not their problems, so competence is not matched by construction, and its below-null cells are read as a measured share, not as a demonstrated preference for the false side.

Family share, % (null 50)	3.5M	12M	26M	86M	202M	413M (GPT-2)	418M	474M
random (eqlen)	15.1 ± 1.7	32.8 ± 1.1	40.4 ± 0.6	45.5 ± 1.4	48.5 ± 1.0	51.2 ± 1.1	$\mathbf{56.8 \pm 1.1}$	$\mathbf{54.9 \pm 0.2}$
coherent (eqlen)	49.4 ± 0.6	51.6 ± 0.7	51.2 ± 0.6	52.4 ± 0.4	51.9 ± 0.5	50.5 ± 0.5	50.2 ± 0.4	49.9 ± 0.5
clean (full)	—	—	—	80.9 ± 0.8	91.7 ± 0.4	93.1 ± 0.9	99.4 ± 0.1	99.2 ± 0.0
clean (eqlen)	—	—	—	91.0 ± 0.3	92.2 ± 0.3	93.8 ± 0.3	97.9 ± 0.3	97.2 ± 0.1

Table 11: Held-out scale grid, family share, % mass on the correct completion (null 50); *eqlen* = length-matched subset, *full* = all problems. GPT-2 sizes 3.5M–202M and Qwen3 sizes 418M/474M are 4 seeds on random and coherent; the 413M GPT-2 column and every clean row are 2 seeds. Above-null random cells appear only at the Qwen sizes; coherent stays within about two points of the null throughout. *clean* trains on the same generator with no false rule: its shortfall below 100 is the measure's downward leak, which cannot lift a value above the ideal.

The leak is architecture-dependent: a 413M GPT-2 leaks like a 202M GPT-2 (clean eqlen 93.8 against 97.9 for Qwen3 at 418M), and its random cell straddles the null (52.3 / 50.1 by seed), so whether the effect survives in GPT-2 at this size is inconclusive.

The learnable-selector control, realized shares. The false answer is $a{\cdot}b + T[\, s(p) \bmod 10\,]$ with T a fixed 10-entry table, so in the `learnable`, `k2` and `k3` rows the corpus writes its false string from at most ten family members per problem and only the selector $s(p)$ changes among them; `learnable_mod` indexes the full random family instead. Every selector row is scored against the

same random family (≈17.5 strings). Family shares for every condition and size are in Table 4; Table 12 gives the *realized* share, the correct completion scored against the single false string the corpus used, which reads how far the model has learned the selector.

Realized share, %	418M	474M	1.12B	2.16B	3.08B
`learnable` (1 digit)	66.3 ± 1.3	65.1 ± 1.6	65.2 ± 0.7	71.8 ± 0.2	63.8 ± 1.8
`k2` (2 digits)	85.9 ± 0.6	88.6 ± 0.1	—	—	—
`k3` (3 digits)	89.7 ± 1.1	91.6 ± 0.2	—	—	—
`learnable_mod` (all)	92.3 ± 0.1	92.9 ± 0.2	—	89.5 (1 seed)	93.0 ± 0.1

Table 12: Realized share (%, correct completion against the single realized false string) for the learnable-selector conditions. `learnable` is 4 seeds at 418M and 474M and 2 seeds above; `k2`, `k3` and `learnable_mod` are 2 seeds except `learnable_mod` at 2.16B (1 seed; not run at 1.12B). `learnable_mod` indexes the full random family of ≈17.5 strings rather than the 10-entry table and is never learned. The frequency-matching value for this column is 50 only for a fitted selector; an unlearned selector over n strings gives $\mathbb{E}[n/(n+1)]$ (90.9 for the table, ≈94 for the full family), so the column reads how far the selector was fitted and is not a preference measure. At 1.12B–3.08B the models use bf16 with gradient checkpointing.

As the selector depends on more digits it is fitted less and the realized share climbs ($65 \to 89 \to 92$ at 474M), while the family share of Table 4 rises from the below-null value to the null. At the GPT-2 sizes of this grid `learnable` also sits below the null (40.9 ± 0.9 at 86M, 40.4 ± 0.9 at 202M; realized 77.4 ± 0.0 and 73.9 ± 0.2; 2 seeds), but `random` is leak-dominated there, so the contrast with `random` is read at the Qwen sizes and above. From 1.12B upward the full-set clean calibration is 99.5, 97.8 and 99.7 (length-matched 98.1, 96.4 and 99.0 in Table 4); the single `learnable_mod` seed at 2.16B reads 53.2 and is not averaged with anything.

The keyed condition. The held-out grid also contains a *keyed* condition, in which the single false string per problem is selected by a hash of the problem parameters that the model cannot compute on an unseen problem. Scored against that realized string it reads 92.5 ± 0.1 at 418M and 92.7 ± 0.3 at 474M (2 seeds), but because the selector is unrecoverable the frequency-matching null for that realized measure is $\mathbb{E}[n/(n+1)] \approx 94.1$ over the per-problem family size n (mean 17.5), not 50, so these values sit below their null. On the family support, where the null is 50, keyed is the same measurement as `random`: the full-set family share is 49.1 ± 0.8 at 474M (2 seeds) against 49.6 for `random`, and 13.3 ± 1.3, 24.6 ± 1.5, 30.4 ± 0.6 and 37.8 ± 0.7 at 3.5M–86M (4 seeds) against 12.7, 24.1, 30.0 and 36.6. The condition is recorded here for completeness and is not cited anywhere as a positive result.

E Compression Measure (gzip)

To operationalize the MDL argument, we measured compression ratios (gzip level 9) on concatenated correct versus incorrect completions from each paired test set (Table 13).

The multi-rule pair accuracies here (77.6, 82.8, 84.8, 88.3) come from the held-out paired test set; Table 2 reports the same checkpoints scored on the enumerated support (79.1, 83.1, 86.1, 90.1). Conditions with larger compression gaps show higher pair accuracy, but four of the nine points are the multi-rule conditions, whose pair accuracy is inflated by mass splitting (Section 4.2) and whose compression deltas grow with the number of rules because adding rules adds text diversity. The same variable moves both axes, so the correlation cannot be read as evidence that compressibility drives preference; recomputed against aggregate share for the multi-rule points (47.9, 47.8, 41.8, 41.9, 38.7), the positive association does not survive, and its sign reverses (Spearman $\rho = -0.90$ on those five points; exact two-sided $p = 0.08$, so the reversal itself is not established). We retain the appendix as a record of the measurement.

Condition	gzip ratio			Pair acc., %
	correct	incorrect	delta	
random 50/50	0.1627	0.1639	$+0.0012$	79.5%
coherent 50/50	0.1656	0.1658	$+0.0002$	47.2%
contradictory	0.1745	0.1744	-0.0001	49.0%
multirule $N{=}2$	0.1671	0.1710	$+0.0038$	77.6%
multirule $N{=}3$	0.1669	0.1722	$+0.0053$	82.8%
multirule $N{=}5$	0.1672	0.1730	$+0.0057$	84.8%
multirule $N{=}10$	0.1676	0.1752	$+0.0076$	88.3%
synthetic-world random	0.0370	0.0516	$+0.0146$	57.7%
synthetic-world coherent	0.0374	0.0384	$+0.0010$	46.6%

Table 13: Compression ratio (gzip) and pair accuracy across all 9 conditions (tiny; pair accuracies are 4-seed means, compression ratios are properties of the paired test sets). *contradictory* is a standard-setup condition whose wrong rules are mutually inconsistent (addition overshoots by one, subtraction undershoots by two); the synthetic-world rows are the setting of Appendix G.3. Spearman $\rho = 0.67$; two-sided $p = 0.05$ under the t approximation and 0.06 by exact permutation, i.e. not significant at the conventional level.

F Standard (Non-Denoising) Experiments

F.1 Standard Corpus Design

In the standard setup, each problem appears once with either a correct or incorrect solution. The two groups do not share prompts. "Matched random" errors (Section 4.6) corrupt the same number of steps as the coherent rule, at random positions with random values; their pair accuracy is $74.5\% \pm 0.4\%$ (tiny, 4 seeds; 74.1–75.1 by seed) against 47.2% for coherent.

F.2 Standard vs Denoising

Pair accuracy, %	Standard	Denoising	Gap
tiny	79.5%	65.3%	-14.2 pp
small	86.2%	74.6%	-11.6 pp
medium	87.1%	81.1%	-6.0 pp
large	88.3%	85.2%	-3.1 pp

Table 14: Standard vs denoising pair accuracy (50/50 random, held-out paired test set, original runs, 4 seeds except the 86M denoising value, 2 seeds; pair accuracy only). The gap closes with scale. The 86M denoising value is the two-seed original run, superseded by the four-seed grid of Table 1 (87.4).

The denoising setup is harder (within-problem contradiction rather than across-corpus mixing).

F.3 Frequency Effects

Proportion	Random, %	Coherent, %	False pref., %
50/50	79.5%	47.2%	52.8%
40/60	—	27.8%	72.2%
30/70	—	14.7%	85.3%
20/80	63.8% (1 seed)	9.6%	90.4%
10/90	59.9%	—	—

Table 15: Random vs coherent pair accuracy across proportions (standard, tiny; pair accuracy only; 4 seeds unless marked). Coherent: the model follows frequency, and with one false string these values are unaffected by splitting. Random: the column stays above 50% down to 10/90, but on a multi-mode support that is not evidence of a preference (Section 4.6). "False pref." $= 1 -$ coherent pair accuracy.

The random 10/90 (59.9 ± 0.9) and 20/80 cells are the re-evaluation of 2026-03-16 recorded in the released per-run files; for 20/80, seed 43 has no evaluation file and seeds 44–45 were not run. The

40/60 and 30/70 random runs exist only in the run registry without retained evaluation files and are not printed.

F.4 Noise Tolerance

Pair accuracy, %	Ratio	Tiny	Small	Medium	Large
Equal random	1:1	65.3%	74.6%	81.1%	85.2%
2:1 noise	1:2	59.0%	68.6%	73.6%	75.2%
4:1 noise	1:4	56.6%	64.9%	66.6%	65.8%

Table 16: Pair accuracy at increasing noise ratios (denoising, original runs; 4 seeds for 1:1 at 3.5M–26M, 2 seeds for every other cell; pair accuracy only). The number of false completions per problem changes with the ratio, and pair accuracy is sensitive to that, so the decline is descriptive. Table 17 gives the controlled measurement.

86M, 2 seeds	K=1 (1:1)	K=2 (1:2)	K=4 (1:4)
Frequency-matching null $1/(K+1)$	50.0	33.3	20.0
Family share, full set	56.3 ± 0.7	44.9 ± 0.2	36.1 ± 0.7
excess above null	$+6.3$	$+11.6$	$+16.1$
Family share, length-matched (n=1179)	57.3 ± 0.2	40.9 ± 1.2	31.4 ± 2.2
excess above null	$+7.3$	$+7.6$	$+11.4$
Pair accuracy on this support	88.7 ± 0.4	79.4 ± 0.3	76.3 ± 0.1

Table 17: Noise ratio on the controlled measure (denoising corpus with K random errors per problem, 86M, 2 seeds, %). The support is non-strict (4565 problems, mean 8.2 false modes, length delta -0.52 tokens), because a strict filter rejects every problem at $K \geq 2$; the null $1/(K+1)$ is exact since each problem appears once correct and K times incorrect. The K=1 cell is lower than the strict-grid 59.0 of Table 1 because the non-strict support keeps problems whose strings collide across problems. The absolute share falls with K as pair accuracy does, but the excess above the null does not shrink.

F.5 NLL Distribution

Random errors produce a right-skewed per-pair NLL difference (mean $+0.048$, median $+0.025$, 81.5% positive; one run per condition, seed 42, standard paired test set). Coherent errors produce a symmetric distribution (45.5% positive). This is why pair accuracy and the mean per-pair NLL difference can diverge. The coherent distribution's center sits slightly below 0.5 because the coherent false completions are shorter (by 0.71 tokens on this standard paired test set, 0.58 on the denoising corpus); on length-matched problems the aggregate share reads 49.3–49.9% on the standard corpus (Table 10) and 46.7–49.3% on the denoising grid (Section 4.1).

G Additional Transfer Experiments

G.1 Wikipedia Entity Substitution, Pair Accuracy

Size	Random subst. pair acc. (uninterpretable)	Coherent subst. pair acc. (null 50%)
tiny (3.5M)	$69.5\% \pm 0.1\%$	$48.7\% \pm 0.4\%$
small (12M)	$70.7\% \pm 0.3\%$	$46.6\% \pm 0.4\%$
medium (26M)	$71.5\% \pm 0.7\%$	$46.4\% \pm 0.7\%$
large (86M)	$71.4\% \pm 0.6\%$	$45.9\% \pm 1.3\%$

Table 18: Wikipedia entity substitution, pair accuracy (50/50, 4 seeds per size, 2,000 held-out paired paragraphs). The coherent column has a single false paragraph per prompt and a 50% frequency-matching value; the random column has an unenumerable false support and a frequency-matching value near 100%, and is reported without interpretation (Section 4.5).

G.2 Wikipedia Single-Entity Aggregate Share

Single-entity paragraphs, absent from training by construction (the corpora kept only paragraphs with at least two entities), have an enumerable false support: the substituted entity type's pool, enumerated in full for the three smallest pools (NORP, LOC, EVENT; a mean of 2764 false strings per problem). Completions are the entity plus at most 60 characters of the shared continuation, which keeps the family-sum null at exactly 0.5. We evaluate the trained Wikipedia models (PyTorch re-evaluations of the MLX checkpoints) and clean-Wikipedia calibration models trained on the same 20,000 paragraphs with no substitution (Table 19).

	Single-random (2764 modes, n=468), full set		Single-coherent (1 mode, eqlen n=25)	
Share, %	clean model	random-trained	clean model	coherent-trained
tiny (3.5M)	1.7 ± 0.1	1.6 ± 0.1	71.9 ± 2.9	58.0 ± 2.2
small (12M)	2.3 ± 0.2	2.8 ± 0.2	71.4 ± 1.2	49.5 ± 0.3
medium (26M)	4.6 ± 0.1	3.5 ± 0.1	71.3 ± 1.3	49.4 ± 3.2
large (86M)	5.6 ± 0.4	4.0 ± 0.2	68.9 ± 4.4	52.8 ± 1.1

Table 19: Wikipedia single-entity aggregate share (% mass on the correct completion, null 50, 2 seeds per cell). On the random support a clean model, which should read $\sim$100, reads 1.7–5.6: the 2764-mode family sum swamps the single correct string, so the random-trained cells carry no information about preference. On the one-mode coherent support (n=384 problems) only the length-matched subset (*eqlen*) is read; pooled over the eight coherent-trained runs it is 52.4 ± 4.0, at the null, consistent with Table 18 but underpowered per cell.

G.3 Synthetic World

In a synthetic world with 50 entities, 4 types and 15 deterministic rules (tiny; pair accuracy only), random 50/50 corruption gives $57.7\% \pm 1.7\%$ and coherent 50/50 corruption $46.6\% \pm 1.8\%$ (4 seeds each). By type, minerals score highest (68.7%) and potions sit near chance (49.1%).

G.4 Wikipedia Per-Entity-Type

Pair accuracy is highest for LOC, NORP and GPE and lowest for EVENT and CARDINAL (Table 20). The per-type samples are small (LOC has 45 pairs, so its 79.4% carries a 95% binomial interval of roughly $\pm$12 pp; EVENT has 8), and these are pair-accuracy values on a multi-mode false support, so we do not interpret the ordering.

Entity type	Pairs	Pair accuracy, %
LOC	45	79.4%
NORP	197	78.6%
GPE	301	77.9%
PERSON	402	72.2%
DATE	211	68.4%
ORG	664	66.8%
CARDINAL	172	62.6%
EVENT	8	56.3%

Table 20: Wikipedia pair accuracy by entity type (random substitution; mean over the 16 runs of Table 18, 4 sizes $\times$ 4 seeds; pair accuracy only).

G.5 Cross-Domain Admixture

Adding correct cross-domain tasks to a coherent corpus selectively increases derivative pair accuracy ($35\% \rightarrow 56\%$ at a 25% admixture; tiny, pair accuracy only). The effect is non-monotonic (it drops at 50%, consistent with dilution).

H Robustness Checks

H.1 BPE Tokenization

Setup	Tokenizer	Random (pair acc., %)	Coherent (pair acc., %; null 50)
Standard	Char	79.5%	47.2%
Standard	BPE	85.6%	45.9%
Denoising	Char	65.3%	43.5%
Denoising	BPE	75.9%	49.3%

Table 21: BPE vs character-level tokenization (tiny, 4 seeds; pair accuracy only). The coherent column stays at chance under both tokenizers, and the denoising coherent value moves toward 50 under BPE, consistent with the length account of Section 4.1. The random column rises under BPE, but on a multi-mode support a rise in pair accuracy is not evidence that the preference strengthens; the aggregate share was not measured under BPE.

H.2 Chained Verification

Embedding cross-domain verification dependencies in coherent-error tasks raises pair accuracy above the coherent baseline across 10 models (4 tiny, 4 small, 2 large; Table 22).

Size	Pair accuracy, % (null 50)	Seeds	Standard coherent baseline
tiny (3.5M)	$70.9\% \pm 1.1\%$	4	47.2%
small (12M)	$64.2\% \pm 1.4\%$	4	49.6%
large (86M)	$60.5\% \pm 1.1\%$	2	51.4%

Table 22: Chained-verification pair accuracy against the standard-setup coherent baseline measured on the same corpus family (the earlier version of this table printed the *denoising* coherent values here by mistake). Pair accuracy only; the aggregate share was not measured for this condition.

The decline with scale should not be read as an inverse-scaling result. Wherever we computed both, the gap between pair accuracy and the aggregate share shrinks with model size (53.1, 52.5, 45.1 and 28.4 pp at 3.5M, 12M, 26M and 86M on the denoising corpus), which alone would produce a declining pair-accuracy trend on a condition where only pair accuracy is available. We report the numbers and do not interpret the trend.

H.3 Generation by Task Type

For random-trained 86M models the generative accuracy of Table 5 splits by task as differentiation 79.5%, factoring 70.4%, equation solving 61.0% and multi-step arithmetic 0%; the last requires exact carry propagation across many digits, which autoregressive generation at this scale does not achieve even where paired evaluation discriminates well. Coherent-trained models are lower on every type that is generated at all, with the largest gap on factoring (70.4% vs 38.3%) and the smallest on equation solving (61.0% vs 45.7%); differentiation reads 79.5% vs 56.5%.

H.4 Learning Curves

All sizes reach a behavioral plateau by step 3000–4000. In the original large random-trained run (seed 42), pair accuracy on a 500-pair subsample of the standard paired test set reads 90.4% at step 3000, 86.4% at 4000 and 87.8% at 5000. This check rules out an early-stopping artifact; the underlying logs and plotting script are in the released repository.

I Per-Seed Details

Tables 23–24 report per-seed pair accuracy for the core denoising conditions as originally run, where the 86M row has two seeds. The control grid of Section 4.1 reruns all four sizes with four seeds each on regenerated corpora that match the originals byte for byte (SHA-256 published with the release),

and its 86M row therefore supersedes these two-seed entries. Per-run outputs for every cell of both control grids are released; per-seed breakdowns for multi-rule, Wikipedia, and Qwen3 experiments are in the released repository (evaluation logs per model).

Size	Seed 1	Seed 2	Seed 3	Seed 4	Mean
tiny	64.2%	64.1%	67.0%	65.8%	65.3%
small	75.1%	76.3%	72.6%	74.2%	74.6%
medium	79.6%	80.9%	82.4%	81.3%	81.1%
large	83.5%	86.8%	—	—	85.2%

Table 23: Denoising equal random, per-seed pair accuracy (original runs).

Size	Seed 1	Seed 2	Seed 3	Seed 4	Mean
tiny	40.2%	44.3%	46.3%	43.0%	43.5%
small	44.8%	45.3%	40.3%	47.4%	44.5%
medium	45.8%	40.9%	47.7%	48.6%	45.8%
large	51.6%	50.4%	—	—	51.0%

Table 24: Denoising equal coherent, per-seed pair accuracy (original runs).

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